

Accessible Learning Platforms: Investigating the Accessibility Standards of ILIAS for Visually Impaired Users

Zeynep Öztürk
FHV Vorarlberg University of Applied Sciences
Austria
zeynep.oeztuerk@students.fhv.at

Dr. Peter Hoffmann
FHV Vorarlberg University of Applied Sciences
Austria
peter.hoffmann@fhv.at

Abstract: Digital accessibility is essential for equitable participation in higher education, yet many learning platforms remain difficult to navigate for visually impaired users. This paper presents one of the first task-based empirical accessibility evaluations of ILIAS, a widely adopted open-source learning management system, applied to a customised university installation adapted to institutional corporate design. A mixed-methods approach combined WCAG 2.1 conformance checks (WAVE tool) with user-centred testing involving screen readers (JAWS), eye tracking, and think-aloud protocols with participants experiencing different degrees of visual impairment. Tasks covered both content navigation and assessment completion. The findings expose persistent barriers – including low contrast, non-semantic structures, inaccessible forms, and limited keyboard operability – revealing a significant gap between formal compliance and real-world usability. By linking branding-related adaptations with authentic user experience, this study offers new perspectives on how institutional design choices intersect with accessibility, thereby advancing the understanding of inclusive digital learning environments.

Introduction

The digital transformation has significantly reshaped education, opening new possibilities for inclusive learning. For individuals with disabilities, accessible digital tools are key to enabling participation in academic, professional, and cultural contexts. However, despite growing awareness, many educators and students still lack the practical skills and guidance to use these tools effectively. Bridging this gap is essential for supporting independent learning and fostering equal participation – objectives that can be realised through platforms built on accessibility and user-centred design principles [1]. While accessible content is important, the underlying architecture of learning platforms is equally critical. Adherence to standards such as the WCAG and WAI-ARIA ensures that digital learning environments are navigable and comprehensible for all users, including those with disabilities. Usability engineering and systematic accessibility testing strengthen this by assessing interactive components such as forums, wikis, and quizzes [2]. Moreover, instructional design plays a key role in accessibility. Studies show that accessible presentations and recordings improve usability and reduce post-production effort. Therefore, authoring tools – such as editors and quiz builders – must comply with the Authoring Tool Accessibility Guidelines (ATAG) to support accessible content creation from the outset [3].

This paper presents a focused accessibility evaluation of ILIAS, a widely adopted open-source learning management system. This study represents one of the first task-based empirical accessibility evaluations of ILIAS conducted on a customised university installation adapted to institutional corporate design. Such branding adaptations, while essential for institutional identity, can unintentionally create accessibility barriers – particularly in colour contrast and layout. By combining WCAG 2.1 conformance checks with user-centred testing involving both learners and a blind instructor using JAWS, the study reveals critical gaps between formal compliance and real-world usability. This dual perspective – branding-related customisation and authentic user experience – makes the contribution unique and provides actionable insights that extend beyond conventional accessibility evaluations of learning platforms.

Problem Statement

Digital accessibility ensures that websites, documents, applications and digital content are usable by all individuals, regardless of personal circumstances [4]. In higher education, this is essential: inaccessible systems limit learning, communication, and participation. Yet, accessibility remains widely neglected – only 4% of the world's top one million websites are accessible [5], and 98% fail to meet WCAG 2.1 standards [6]. At Vorarlberg University of Applied Sciences, where 1,627 students are enrolled [7], it is imperative that platforms such as ILIAS address diverse accessibility needs, including visual, cognitive, and physical impairments. This study is driven by a commitment to digital equity and was developed in collaboration with accessibility experts and visually impaired educators. Conducted in a dedicated usability lab and approved by an ethics board, it aims to identify practical accessibility barriers in ILIAS and promote more inclusive learning environments. Although the digitalisation of education presents significant opportunities, learning management systems (LMS) often fall short in supporting users with disabilities [8]. Technical compliance alone is insufficient – real user interaction is essential to uncover usability gaps [1]. Many platforms fail to meet legal obligations such as those outlined in the European Accessibility Act [2], and educators lack accessible, practical and low-threshold resources to support implementation. A dual-method approach – combining automated audits and user testing – is thus necessary to bridge the gap between formal standards and genuine inclusion.

State of Accessibility in E-Learning

This section reviews related work on accessibility in digital learning environments and situates the present study within this context. The understanding of disability has evolved from a deficit-oriented medical model to a social model that emphasises the role of environmental and systemic barriers in limiting participation [9], [10]. This shift reframed accessibility as a design responsibility rather than a personal limitation. Today, accessibility is defined as the inclusive design of environments, services, and technologies to ensure equal usability for all users, regardless of sensory, physical, or cognitive differences [11].

In digital contexts, accessibility is operationalised through the WCAG developed by the W3C [12]. The current WCAG 2.1 standard expands upon earlier versions by addressing the needs of users with low vision, cognitive impairments, and mobile device users [13], [12]. Its four guiding principles – Perceivable, Operable, Understandable, and Robust (POUR) – provide a comprehensive framework for inclusive design [12], [14]. However, even full compliance with Level AA, the prevailing legal standard, does not necessarily ensure functional usability [14]. In Austria, accessibility is mandated by national and EU legislation, including EU Directive 2016/2102 and the forthcoming European Accessibility Act, which will extend obligations to the private sector by 2025 [15], [16]. Despite this, evaluations reveal that around 36% of Austrian public websites still fail to meet accessibility standards [17]. In parallel, digital inclusion has become a key educational principle, promoting learning environments that proactively accommodate diversity. Frameworks such as Universal Design for Learning (UDL) support this approach [18], [19]. E-learning platforms, including LMSs and massive open online courses (MOOCs), play a central role in delivering accessible content [20], [21], increasingly integrating assistive technologies such as screen readers, Braille displays, and keyboard navigation [13], [22]. Prior research has further highlighted that instructional methods originally developed for learners with typical visual acuity are often applicable to those with visual impairments, yet structural and interface-level barriers persist and necessitate targeted accessibility interventions [23].

Complementing this, Goodman's seminal work on the universal nature of meaning making through reading and writing underscores that both sighted and visually impaired learners rely on similar language cueing systems to build knowledge from textual input [24]. These perspectives reinforce the notion that accessibility must not only be seen as technical compliance but also as the facilitation of universally shared cognitive processes through inclusive design.

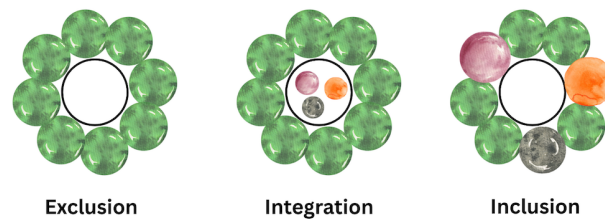


Figure 1. Conceptual representation of exclusion, integration, and inclusion, inspired by [25]

ILIAS, a widely used open-source LMS, supports modular content delivery, collaboration tools, and data protection through self-hosting [26]. While WCAG 2.1 compliance is progressing, challenges remain – particularly regarding screen reader consistency and keyboard navigation [27]. The platform's SIG Accessibility group continues to address these issues through community-driven development initiatives [26]. Prior work largely operationalises accessibility through WCAG/WAI-ARIA and code-level audits, but fewer studies examine authentic interaction of visually impaired users with LMS in task-based scenarios. Evidence for ILIAS specifically is limited. This study addresses this gap by combining a WCAG-based audit with empirical testing involving participants with varying degrees of visual impairment, to contrast formal conformance with real-world usability.

The Study / Methodology

To evaluate the accessibility and usability of the ILIAS platform for visually impaired users, this study employed a mixed-methods design combining technical conformance analysis with empirical user testing. This dual approach enabled the identification of both WCAG 2.1 [12] compliance issues and real-world usability challenges, offering a comprehensive perspective on accessibility from technical and experiential viewpoints. The methodology was developed in consultation with the ILIAS SIG Accessibility group and informed by established principles in inclusive user experience research [28], [29].

Workflow-Based Scenario Design

Given the complexity and modular structure of ILIAS, a complete audit of all features would be impractical. Following recommendations from the SIG Accessibility, the study adopted a scenario-based approach focused on a representative, pedagogically relevant user workflow: navigating a structured content page and completing an associated online test. These tasks reflect common learning activities and rely solely on ILIAS-native components, making them ideal for isolating platform-specific accessibility barriers [30]. A private test environment was created within a dedicated ILIAS group, featuring role-based folders for students and instructors. Both roles accessed identical learning objects to ensure content consistency and to isolate interface-based role differences.

The content page included structured text, lists, tables, links, and images with alt attributes, simulating a typical lecture module. Known accessibility-critical formatting styles were deliberately applied to assess their impact on screen reader compatibility and keyboard navigation [31]. The online test included six native question types – multiple choice (single answer), cloze, matching, ordering, image map, and essay – to evaluate the accessibility of ILIAS's assessment tools. The test measured whether users could perceive, understand, and interact with the various question formats using assistive technologies, without assessing content knowledge.

Technical Evaluation Using WAVE and Firefox Developer Tools

A systematic accessibility evaluation was conducted using WAVE (Web Accessibility Evaluation Tool) [31] and Firefox Developer Tools [32], focusing on WCAG 2.1 Level AA conformance [12]. WAVE identified issues related to semantic structure, ARIA roles, keyboard accessibility, and contrast ratios. Firefox's accessibility inspector complemented this by providing a detailed accessibility tree and enabling simulations of various colour vision deficiencies, including protanopia, deuteranopia, tritanopia and achromatopsia [33]. Simulated visual perception filters in Firefox Developer Tools were employed to document how colour-based cues appeared under different types of colour vision deficiency.

Empirical Testing with Visually Impaired Participants

To complement the technical audit, user testing was conducted with six participants experiencing varying degrees of visual impairment, including one blind instructor. Ethical approval was obtained from the university's Research Ethics Committee, and informed consent was secured through accessible formats – including a Braille version and a 3D-printed signature guide for blind participants [34].

While the number of participants may appear limited, this decision was deliberate to enable in-depth engagement with each individual. Prior usability research has shown that small participant numbers are sufficient to uncover the majority of critical issues, with Nielsen demonstrating that as few as five participants can reveal approximately 77–85% of usability problems [35]. In this study, six participants were deliberately chosen to exceed this threshold, ensuring methodological rigour while capturing a broader spectrum of visual conditions. Participants were observed in the UX Lab at Vorarlberg University of Applied Sciences, which is equipped with Tobii Pro Lab for gaze tracking and behavioural analysis (see Figure 2) [36]. The setup consisted of a single-room layout with the moderator positioned diagonally to enable verbal support without visual interference [37]. Eye tracking was employed with participants who had residual vision to record fixations, navigation patterns, and hesitation points. A standard five-point calibration ensured measurement precision [38].

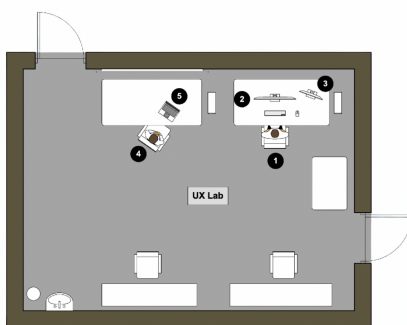


Figure 2. Setup for Eye-Tracking Tests

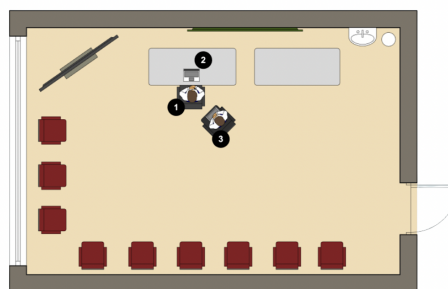


Figure 3. Setup for Non-Visual Test

For the blind participant, a non-visual testing protocol was implemented using their personal laptop and a customised JAWS screen reader configuration (see Figure 3) [29], [39]. This setup adhered to best practices in accessibility testing, allowing participants to interact within a familiar environment using their preferred assistive technologies [40], [41]. Verbal instructions followed the Partial Concurrent Think-Aloud protocol to minimise auditory overload [42].

Thinking Aloud and SUS Protocols

All participants were instructed to verbalise their thoughts while interacting with the platform, following the Thinking Aloud protocol. This method provided insights into users' mental models, decision-making processes, and the cognitive barriers encountered during navigation [43]. Upon completing the test scenario, each participant engaged in a semi-structured interview based on Nielsen's usability heuristics and completed a modified System Usability Scale (SUS) questionnaire, adapted to reflect accessibility-specific concerns [44]. This triangulated approach enabled the collection of both behavioural and subjective usability data, enhancing the validity and robustness of the study. By integrating automated WCAG 2.1 analysis with qualitative user insights, the methodology offered a comprehensive perspective on the barriers and enablers of accessibility within ILIAS. This dual perspective forms the empirical foundation for the recommendations presented in the following chapter.

Findings

To assess the practical accessibility and usability of the ILIAS platform for visually impaired users, this study adopted a mixed-methods approach that integrated both quantitative metrics and qualitative insights. Data collection included eye-tracking visualisations, think-aloud protocols, structured observation and the SUS. Participants were divided into two cohorts based on visual ability: five individuals with partial vision (TP01–TP05) and one congenitally blind participant (TP06), allowing for differentiated analysis across the visual impairment spectrum (see Table 1) [45].

Participant ID	Left Eye (Sph)	Right Eye (Sph)	Other Visual Conditions
TP01	-2.00	-4.50	Astigmatism
TP02	-0.50	-1.25	Mild Astigmatism
TP03	-5.00	-5.00	Mild Strabismus
TP04	-0.50	-0.25	Astigmatism
TP05	-4.50	-4.75	Extraocular Muscle Strain
TP06	N/A	N/A	Congenitally blind

Table 1. Visual Characteristics and Diagnosed Conditions of Participants

Task Performance and Interaction Behaviour

Participants were divided into two cohorts: five partially sighted users (TP01–TP05) and one congenitally blind instructor (TP06). All participants engaged with a predefined scenario involving content navigation and the completion of a structured online test.

Eye-tracking data revealed task-level variations in cognitive load. Launching the test (Task 3) required the shortest time on average ($\bar{x} = 45.6$ s), whereas reading the content page (Task 2) was the most time-consuming ($\bar{x} = 394.6$ s). This suggests elevated cognitive effort due to a dense layout and limited visual guidance.

Among the test items, drag-and-drop formats such as ordering (Q2) and matching (Q5) posed substantial interaction barriers, supporting existing critiques of non-accessible input modalities. In contrast, multiple-choice and single-text questions were completed with minimal difficulty.

Participant TP05 showed significantly lower performance across all tasks, which can be attributed to unmitigated visual fatigue caused by diagnosed extraocular muscle strain and a fixed screen distance. This highlights that visual acuity alone is not predictive of accessibility success.

Visual Interaction Data (Tobii Pro Lab)

The Tobii Pro Lab recordings allowed a detailed analysis of gaze behaviour. Applying the I-VT (Attention) filter, participants showed frequent re-fixations and delayed saccades on low-contrast elements, such as ILIAS-native paragraph headings and emphasis styles, which did not meet WCAG 2.1 contrast ratios (e.g., 1.67:1 for blue-on-white text). Key controls like the “Start Test” button often escaped initial attention, indicating a misalignment between visual salience and functional relevance.

Fixation metrics – such as time to first fixation, total fixation duration, and fixation count – were used to evaluate Areas of Interest (AOIs). Links with sufficient contrast on paper (e.g., white-on-orange, #ED6E4C) were still overlooked or discovered late, suggesting perceptual issues not captured in static audits. Longer fixations and reduced saccades on essay questions and dense layouts indicated elevated mental effort. These behavioural signals, when triangulated with verbal data, revealed that technically compliant interfaces may still present functional barriers.

Non-Visual Interaction: Screen Reader User (JAWS)

The blind participant (TP06) completed the same scenario as sighted users, relying entirely on auditory feedback via JAWS. The session was conducted in a distraction-free setting and adapted to support non-visual autonomy, including accessible consent materials and a personalised screen reader configuration. All instructions were given verbally, and the participant demonstrated confident navigation based on high digital literacy and familiarity with the environment.

As expected, task durations were longer due to linear interface parsing (e.g., Q7 = 161 s). Complex or inconsistently structured elements – such as drag-and-drop formats and poorly grouped radio buttons – resulted in hesitation, inefficiency, or skipped items. These barriers stemmed from missing semantic cues and inadequate screen reader support. Despite encountering several obstacles, TP06 achieved a 69.70% test score. This outcome illustrates that capability alone does not guarantee access; rather, inclusive design determines operability. With more consistent markup and accessible interaction logic, full task completion would likely have been possible.

Observational Insights and Verbal Reports

Think-aloud data uncovered consistent patterns: low contrast, ambiguous button placement, and small font sizes were frequently cited as disorienting. Tasks requiring high visual scanning effort led to fatigue described metaphorically as “looking through water.” These perceptions corroborate studies on visual strain and layout complexity. Drag-and-drop interactions emerged as a critical bottleneck, especially in high-stakes settings like assessments, where bypassing is often impossible. The absence of fallback options contradicted inclusive design principles and forced dependence on external assistance, undermining user autonomy. Positively, participants praised cleanly structured sections, consistent layouts, and logical reading order. These elements, when present, improved orientation and reduced cognitive burden.

System Usability Scale (SUS)

SUS scores ranged from 37.5 to 77.5, with a group median slightly above the standard usability benchmark of 68 [44]. While TP01–TP03 reported positive usability experiences, TP05 and TP06 rated the platform lower, in alignment with their observed navigation difficulties. These results support the broader finding that technical conformance alone does not ensure perceived usability [46].

SUS thus functioned as a diagnostic tool, exposing both user satisfaction and latent frustration – particularly among individuals facing high visual complexity or unclear interaction logic. When triangulated with eye-tracking data and verbal protocols, these scores underscore the critical importance of perceptually inclusive design.

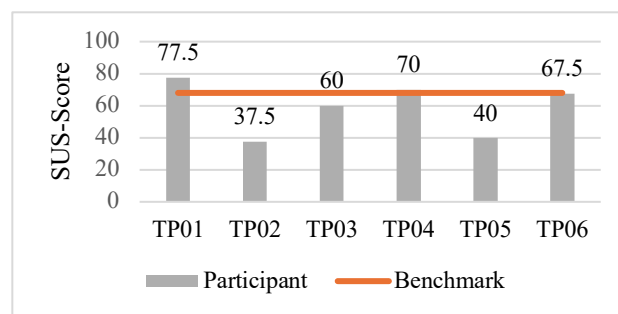


Figure 4. SUS Scores Compared to Benchmark

Empirical Design Guidelines

The findings suggest that WCAG 2.1 conformance alone does not ensure accessible user experiences. True accessibility must address perceptual, motor, and cognitive needs through user-centred design.

First, insufficient colour contrast – such as light text on light backgrounds – caused readability issues, particularly for users with visual impairments. While certain colour combinations may align with institutional branding, they should always be verified for sufficient contrast according to accessibility standards.

Second, text that is visually highlighted but lacks proper semantic structure can hinder both screen reader interpretation and visual scanning. Instead of relying solely on font styles or colours to indicate hierarchy or emphasis, semantic HTML elements and ARIA roles should be used to convey meaning accessibly.

Interactive components – such as drag-and-drop tasks or image-based quizzes – frequently posed barriers due to missing keyboard support and insufficient screen reader feedback. To ensure usability, each interactive activity should have an accessible, functionally equivalent alternative.

Inconsistent placement of navigation controls (e.g., buttons for submitting or progressing through content) often led to orientation delays, especially for users relying on magnifiers or auditory cues. Clear, predictable layouts are essential to minimise cognitive effort and improve overall usability. Additionally, non-descriptive link labels (e.g., "click here") were difficult to interpret in screen reader contexts. Replacing such links with clear, meaningful text improves orientation and aligns with WCAG 2.4.4.

Importantly, accessibility and institutional design goals can coexist. With thoughtful planning, interface elements can be styled in ways that reflect an organisation's visual identity while still meeting accessibility requirements. In summary, accessibility must be approached as an inclusive design process that anticipates user diversity from the beginning – not as a checklist to complete post-development.

Conclusion

This paper presented a comprehensive evaluation of the ILIAS learning platform with a focus on accessibility for users with visual impairments. By combining automated WCAG 2.1 assessments with empirical user testing, the study demonstrated that technical compliance alone is insufficient to guarantee genuine accessibility.

While ILIAS incorporates valuable structural features such as semantic HTML and screen reader support, persistent barriers remain – including low visual contrast, non-semantic formatting, and interaction types that are not operable without vision. These findings emphasise that accessibility is not a checklist of technical requirements but a continuous design responsibility that must anticipate diverse learner needs.

A distinctive contribution of this study lies in providing one of the first task-based empirical accessibility evaluations of ILIAS in a university-specific installation adapted to institutional corporate design, combined with authentic user experiences, including those of a blind instructor using JAWS. This dual perspective revealed how institutional branding decisions and lived user interactions intersect to shape accessibility in practice. The insights have been shared with developers and accessibility advocates to support sustainable improvements and encourage inclusive design decisions in higher education.

Future work should expand this foundation in several directions. Comparative studies involving visually impaired and non-impaired users could clarify which challenges are universal and which are disability-specific. Longitudinal research may examine whether incremental design changes – such as improved colour contrast, semantic structuring, or accessible alternatives to drag-and-drop tasks – translate into measurable usability gains over time. Extending the scope beyond visual impairments to cognitive and motor disabilities would further broaden the inclusivity perspective. Finally, embedding accessibility into institutional practice – through ATAG-compliant authoring tools, systematic accessibility testing, and instructor training – would ensure that improvements are sustainable and not limited to isolated interventions.

In conclusion, accessibility in digital education must be recognised not as an optional add-on but as a fundamental commitment to equity, dignity, and opportunity. By exposing the gap between compliance and real-world usability, this study delivers actionable evidence for creating digital learning environments that are not only legally compliant, but genuinely inclusive for all learners.

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